

LEARNING MANAGEMENT SYSTEM

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NEXL : ELEVATE YOUR LEARNING

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Project Introduction

E-learning has become a key part of modern education, driven by increasing internet penetration, widespread smartphone adoption, and the shift towards flexible and hybrid learning models. The COVID-19 pandemic accelerated the transition to online learning worldwide, including Nepal, where educational institutions relied heavily on platforms like Zoom and Google Meet to continue education. Studies show high LMS adoption in Nepal, with approximately 83% of students having internet access and around 63% relying primarily on smartphones for learning.

Despite this growth, the commercialization and accessibility of online courses remain limited due to high fees, poor integration of local payment gateways, and lack of interactive learning features. Most international LMS platforms fail to address Nepal-specific requirements, making course monetization difficult for educators and expensive for learners.

To address these gaps, NEXL is a full-stack, web-based Learning Management System tailored specifically for Nepal. It empowers independent educators to create, manage, and monetize their courses using local payment gateways, while providing students with an affordable, interactive, and gamified learning experience. NEXL emphasizes multi-format content delivery, AI-driven learning support, and sustained learner motivation, making it a locally relevant and sustainable solution for Nepal's growing e-learning ecosystem.



Problem Statement

Key Gaps in Existing Systems-Global platforms

- ▶ Global e-learning platforms are expensive and lack localized features, limiting access for Nepali users.
- ▶ International platforms do not support Nepal-specific payment gateways (e.g., eSewa, Khalti), making course purchases difficult.

Local Nepali platforms (Mero School, Neema Academy, etc.)

- ▶ Existing local LMS platforms fail to provide modern, interactive, and user-friendly learning experiences for Nepali learners.
- ▶ Educators in Nepal struggle to monetize their courses due to poor platform support and limited audience reach, while students face affordability and accessibility barriers.



Objectives

- ▶ To design and develop a modern, affordable, Nepal-focused web-based LMS that improves accessibility, interactivity, and sustainability in online education.
- ▶ To integrate secure, low/zero-commission local payment gateways (eSewa, Khalti) for easy and affordable course enrollment.
- ▶ To provide instructors with flexible pricing, coupons, free previews, mixed content models, live streaming, quizzes, assignments, and student performance tracking tools.
- ▶ To deliver AI-enhanced learning with personalized recommendations, lecture summarization, interactive tools (chat, forums, tutoring), badges, rankings, ratings, and mobile/low-bandwidth optimized responsive design.



Methodology and Requirements

Development Approach

- ▶ Iterative & Incremental

Functional Requirements

- ▶ Course management
- ▶ Local payment (eSewa/Khalti)
- ▶ Quizzes & badges
- ▶ Chat & recommendations

Non-Functional Requirements

- ▶ Responsive
- ▶ Secure
- ▶ Scalable

Tech Stack

- ▶ Frontend: React.js
- ▶ Backend: Node.js + Express
- ▶ Database: MongoDB
- ▶ Others: Socket.io (chat), JWT, Cloudinary (files)



▶ Course Recommendation Algorithm

Content-based recommendation using similarity matching on course content and user preferences.

Usage in NEXL: Delivers personalized course suggestions based on user viewing history and course content, increases engagement & discovery.

▶ Content Summarization Algorithm:

AI-based abstractive text summarization to generate concise summaries of lectures and notes.

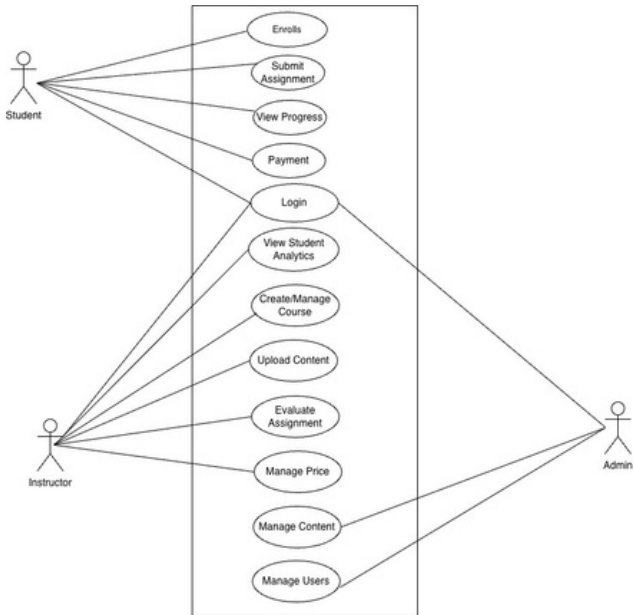
Usage in NEXL: Generates quick, clear summaries of video lectures and notes for faster understanding

▶ Search Algorithm

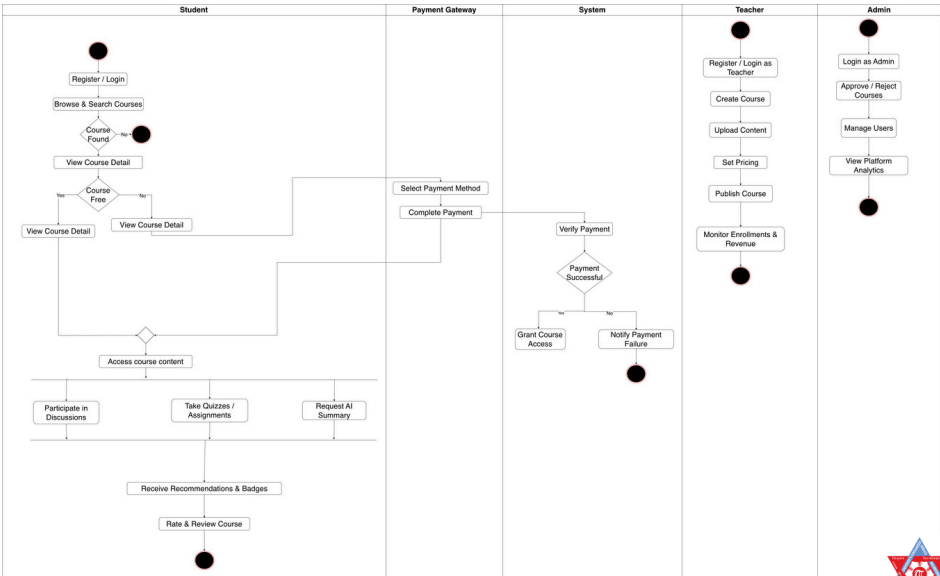
- Course Embedding (run once when teacher creates/updates course)
- User Profile Vector (average of embeddings of recently viewed/enrolled courses)
- Similarity Search (cosine similarity over all published courses)
- Return Top-K recommendations (e.g., top 5)



USE CASE DIAGRAM



ACTIVITY DIAGRAM



GANTT CHART



FIG : TIMELINE OF PROJECT



EXPECTED OUTCOMES

▶ **Seamless & Affordable Payments :**

Integration with eSewa, Khalti etc., enables low or zero commission payments , making course enrollment affordable for students from diverse economic backgrounds.

▶ **AI-Enhanced Learning Features :**

Includes personalized course recommendations, AI-powered lecture and note summarization, and real-time interaction tools such as chat, forums, and one-on-one tutoring.

▶ **Production-Ready & Scalable System :**

A fully tested, cloud-hosted prototype with complete documentation, ready for live demonstration and designed for future scalability and feature expansion.



Thank You

I appreciate your time and attention.
Please feel free to ask any questions or share your
feedback.

